MinPPI

Min-Marginal Power Indices for Multi-Objective Feature Selection

Context-dependent harm in feature selection

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Average importance and contextual harm

Can adding one feature make an acceptable model fail in a particular context?

Average attribution

How does the feature contribute across many feature subsets?

MinPPI

Does at least one subset fall below the utility quota when it is added?

A coalition is a subset of features. Harm depends on the model, split, utility and quota.

A concrete witness of harm

Constructed five-feature game



Marginal contribution: 0 − 1 = −1

The pair of evaluated coalitions certifies the harmful context.

The exact MinPPI definition

$$v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha U_{full}]$$

$$MinPPI_i = \min_{S \subseteq N \setminus \{i\}} [v_\alpha(S \cup \{i\}) - v_\alpha(S)]$$

U is validation utility. α sets the quota relative to the positive full-model utility.

−1

Harmful

At least one coalition flips from acceptable to unacceptable.

0

Neutral at the minimum

No harmful coalition and no universally decisive effect.

+1

Universally decisive

Every coalition flips from unacceptable to acceptable.

Finite sampling can miss harmful witnesses

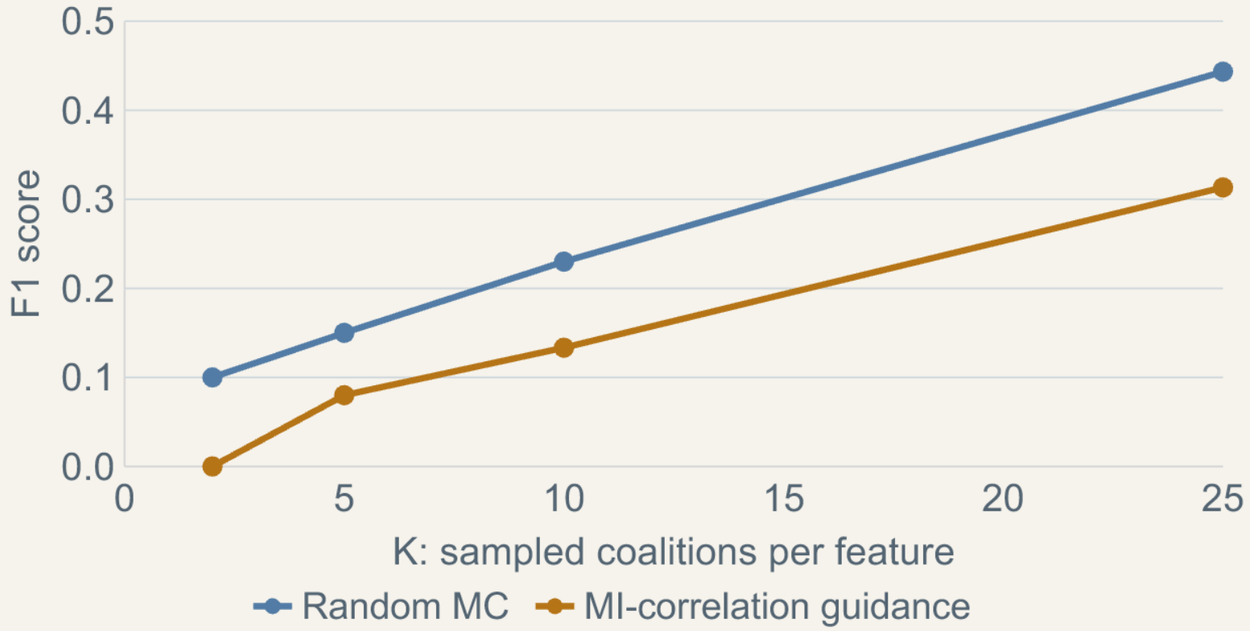
Observed −1

An evaluated coalition certifies the harmful threshold crossing.

Sampled 0 or +1

An unseen coalition may still contain a harmful witness.

Abalone: harmful-label recovery



Five-seed means. F1 combines precision and recall.
Ground truth: exact enumeration of 8 features.
Exact and hybrid recover every harmful label (F1 = 1).

Witness search and the removal decision

01 **Freeze the rule**

Fix the model, split, utility, quota and validation tolerance.

02 **Collect removal candidates**

Use training and validation to find features with an observed −1.

03 **Test joint removal on validation**

Delete all candidates only if the loss stays within tolerance. Otherwise, delete none.

04 **Evaluate on held-out data**

Measure the frozen decision after feature selection is complete.

Allowed validation loss: 0.005 for classification, 0.010 for regression.
The gate cannot guarantee held-out non-inferiority.

Experimental scope and data integrity

Six retained benchmarks, five frozen seeds, one XGBoost family

Train 64% Validation 16% Held-out test 20%

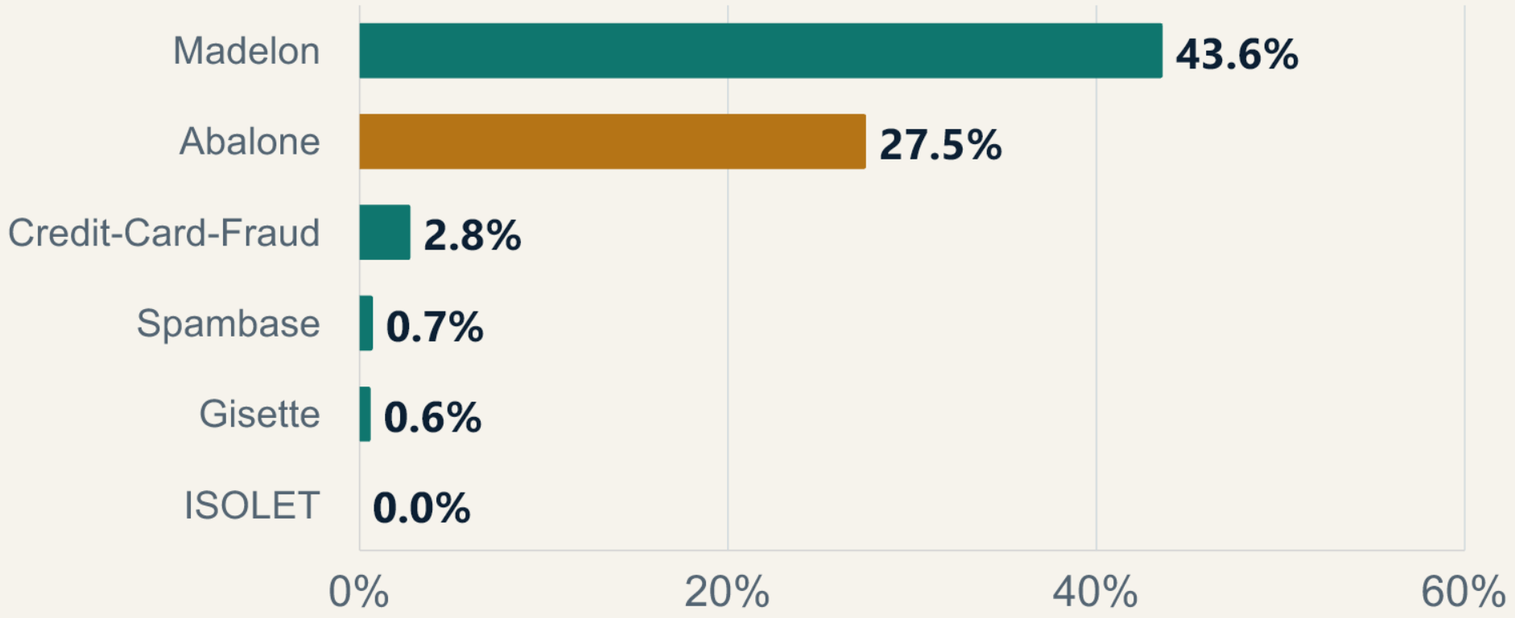
Dataset	Native features	Analysis cap	Utility
Spambase	57	57	ROC AUC
Madelon	500	100	ROC AUC
ISOLET	617	100	OVR AUC
Gisette	5,000	100	ROC AUC
Credit-Card-Fraud	29	29	ROC AUC
Abalone	8	8	R^2

Training-only variance filtering and mutual-information selection determine the analyzed features.

An integrity audit excluded Communities & Crime post hoc for target leakage.
All reported tables and figures use the six remaining datasets.

Removal outcomes depend on the dataset

Mean share of analyzed features removed



Madelon

ΔU = +0.00469

Largest average compression with a small utility increase.

Abalone

ΔU = −0.00565

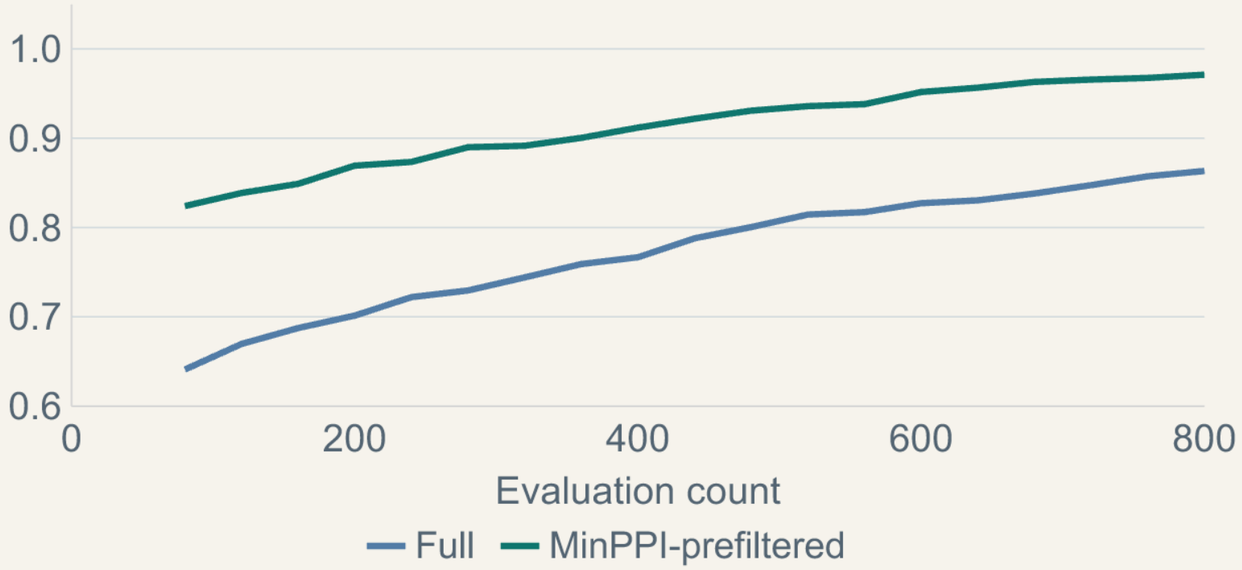
Compression carries a held-out utility cost.

Five-seed means at α = 0.95. ΔU compares removal with the full model.
Compression varies strongly across seeds, especially on Madelon and Abalone.

Search results at the same evaluation budget

NSGA-II balances feature count and utility, using 800 evaluations per variant

Madelon: mean validation hypervolume



Mean held-out hypervolume change

Dataset	ΔHV
Madelon	**+0.09515**
Credit-Card-Fraud	+0.00535
Abalone	+0.00021
ISOLET	0.00000
Gisette	−0.00191
Spambase	−0.00378

HV summarizes the trade-off. Higher is better.

Average ΔHV: +0.01584. No significant improvement after Holm correction (p = 0.4375).

Five-seed means. ΔHV = prefiltered minus full. The study makes no runtime acceleration claim.

Conclusion and further research

MinPPI ties a harmful-feature label to a concrete threshold-crossing witness.

A fixed validation rule decides whether to use the candidate set.

What the evidence supports

Selective compression in this campaign.
The clearest descriptive gains occur on Madelon.
Effects depend on the dataset and seed.

Further research

Improve witness-search coverage.
Replicate across model families and domains.
Strengthen leakage screening.